Code No: 158DA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, July/August - 2022
BASICS OF POWER PLANT ENGINEERING
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) What are different methods used for collection of the dust before sending the flue gas through chimney? Explain them with suitable diagrams.
b) Draw the layout of modern coal power plant and explain its working. [8+7]

2.a) Write a short note on binary cycles and cogeneration systems.
b) Draw and explain the working of Rankine cycle. [8+7]

3.a) Draw the line diagram of gas turbine power plants and explain its components.
b) Explain the closed cycle gas turbine used in the power plants. [8+7]

4.a) Explain in detail about integrated gasifier based combined cycle (IGCC) systems.
b) What are the advantages, disadvantages and limitations of gas turbines? [8+7]

5.a) Explain the construction and working of nuclear power plant with a layout.
b) With the help of a neat sketch, the construction working of a boiling water reactor (BWR). [8+7]

6.a) Describe with the help of a neat sketch, the construction working of a pressurized water reactor. What are the advantages and disadvantages?
b) List out various safety measures to be taken in nuclear power plants. [8+7]

7.a) Draw the line diagram and explain the working details of hydro power plant giving salient points.
b) How to make use of the tides for power generation based on their capacities? Explain the principle of operation. [8+7]

8.a) What is the impact on the environment and human health for the effluents released from the thermal power plants? Explain how to control them.
b) Explain the significance of: i) Load factor ii) Diversity factor iii) Plant capacity factor iv) Plant use factor. [8+7]

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